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A comprehensive review of AI for food recognition and nutrient estimation with an obese children perspective

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Abstract

Background Image-based dietary assessment (IBDA) promises lower burden than recalls but its real-world performance, especially in children, remains uncertain. Objective: To map the IBDA knowledge landscape and synthesize evidence on accuracy, usability, and translational readiness for pediatric obesity prevention and care.

Methods We searched the Web of Science Core Collection (2015–October 2025) for English-language IBDA studies; conducted science mapping (co-citation, thematic/strategic diagrams, temporal keyword evolution); and performed narrative synthesis across recognition, portion/volume estimation, and nutrient inference, emphasizing free-living validation and pediatric subgroups.

Conclusion Publications have grown and coalesced around dietary assessment, nutrition, and validity. Deep-learning and multimodal advances yield strong recognition and feasible portion/energy estimation on curated datasets, yet performance degrades in free-living use due to domain shift, long-tail foods, and error accumulation. For pediatric deployment, clinically useful outputs should prioritize meal-pattern signals and high-leverage categories, communicate uncertainty, and integrate caregiver-compatible workflows, privacy protections, and fairness audits. Pediatric-first validation against appropriate gold standards and benchmarks reflecting children's food ecologies is needed to translate methodological gains into dependable clinical and public-health tools.

Keywords Image-based dietary assessment, Computer vision, Deep learning, Portion/volume estimation, Nutrition assessment

1 Introduction

Childhood obesity has emerged as one of the most pressing public-health challenges of the twenty-first century. Global pooled analyses report a sustained rise in obesity among school-aged children and adolescents across most countries since 1990, with obesity now more prevalent than thinness in a majority of nations, while the overall "double burden" of malnutrition remains high in many regions [1]. Complementing these estimates, the World Health Organization (WHO) highlights obesity's long-term health sequelae

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and its multifactorial etiology, underscoring the urgency of scalable prevention and management strategies beginning in childhood.

Effective dietary assessment is foundational for both prevention and treatment of pediatric obesity, yet traditional methods—24-h recalls, food records, and food-frequency questionnaires—are burdensome and prone to recall and portion-size errors, especially in youth. Systematic reviews find that children with overweight/obesity frequently misreport energy intake relative to doubly labeled water, with under-reporting particularly prominent among adolescents [2]. In clinical guidance, the American Academy of Pediatrics (AAP) emphasizes comprehensive, family-centered management but acknowledges that reliable, low-burden tools to measure intake remain a bottleneck to precision care [3].

Advances in computer vision and mobile sensing have catalyzed a shift "from memory to measurement," enabling image-based dietary assessment (IBDA) workflows that transform meal photos into structured estimates of foods, portions, and nutrients. Recent reviews delineate a pipeline—segmentation, recognition, portion/volume estimation, and nutrient mapping—that increasingly leverages deep learning and large public datasets [4, 5]. Compared with manual records, IBDA can reduce respondent burden and capture contextual information (co-occurring items, leftovers), though it introduces new error sources and still requires careful validation.

Food recognition has benefited from the availability of benchmarks such as Food-101 (101 classes, 101k images) and the UEC-FOOD series (100/256 classes, with bounding boxes and segmentation variants), which capture diversity in cuisines and presentations [6, 7]. Beyond category labels, multimodal corpora like Recipe1M align images with ingredient lists and instructions, enabling vision-language modeling and ingredient-level reasoning that are directly relevant to nutrition estimation [8].

Translating "pixels to calories" hinges on portion and volume estimation. Early systems used geometric priors, fiducial markers, and multi-view reconstruction; more recent approaches apply single-image depth cues and learned shape priors to approximate volume before mapping to energy and nutrients via food composition databases [5, 9]. Pioneering work from the TADA project demonstrated mobile, image-assisted dietary assessment in adolescents, validating the feasibility of pre-/post-meal image capture and automated analysis in free-living conditions [10]. Despite progress, error accumulation across recognition and volume stages, domain shift across cuisines, and mixed dishes remain active challenges for reliable nutrient estimation.

For children with obesity, the design space tightens. Compared with adults, children's eating occasions are more frequent and unscheduled, foods are often packaged/snack-like or school-provided, and parents/caregivers may serve as proxy reporters. These factors exacerbate missed-event and portion-size errors and magnify under-reporting biases documented in validation studies [2]. Moreover, pediatric care demands heightened attention to privacy, assent/consent, and equitable model performance across socioeconomic and cultural contexts that are under-represented in standard food datasets. Aligning with AAP guidance, systems intended for clinical or community use must be interpretable, minimally burdensome, and integrated into family-based behavior change.

Accordingly, this review does not aim merely to summarize recent advances in AI-based food recognition and nutrient estimation. Instead, it evaluates the translational

potential of image-based dietary assessment technologies for pediatric obesity prevention and care. To do so, we couple a science-mapping analysis of the recent literature with a focused narrative synthesis of the end-to-end IBDA pipeline, including segmentation and recognition, portion or volume estimation, and nutrient inference. Our goals are to quantify the field's growth and conceptual structure, examine how benchmark performance compares with free-living applicability, and identify the methodological, ethical, and validation issues that must be addressed before these tools can be used more meaningfully in child-centered public-health and clinical settings.

2 Research questions

RQ1 What are the publication trends, knowledge structure, and thematic evolution of research on AI-based food recognition and nutrient estimation?

RQ2 What are the main technical approaches used in image-based dietary assessment, including food recognition, portion/volume estimation, and nutrient inference?

RQ3 How do these methods perform across curated benchmark datasets versus free-living or real-world settings?

RQ4 What are the major translational challenges and future directions for applying these methods to childhood obesity prevention and care?

3 Materials and methods

3.1 Study design

We employed a two-phase approach. First, we conducted a bibliometric mapping of the literature on AI-based food recognition and nutrient estimation based on records retrieved from the Web of Science Core Collection, Scopus, and PubMed, using a predefined search strategy and PRISMA-style identification, screening, and eligibility procedures to build a clean corpus for science mapping. Second, informed by the bibliometric results, we performed a focused narrative review of representative subdomains, including food recognition, portion or volume estimation, and nutrient inference, to connect high-level trends with practical research directions (Fig. 1).

3.2 Search strategy

A systematic literature search was conducted across the Web of Science Core Collection, Scopus, and PubMed to identify studies related to AI-based food recognition and nutrient estimation. The final search was completed on 1 October 2025, and the search window covered publications from 1 January 2015 to 1 October 2025. To ensure comparability across databases, the search strategy was organized around three concept groups: food or meal imagery, dietary or nutrient assessment tasks, and artificial intelligence or computer-vision techniques. Because indexing rules and field tags differ across databases, equivalent Boolean expressions were adapted for each platform.

The core search logic was based on the following Boolean structure:

("food image*" OR "meal image*" OR "food photograph*" OR "dietary image*" OR "image-based dietary assessment" OR "mobile food record") AND ("food recognition" OR "food classification" OR "food detection" OR "food segmentation" OR "portion size

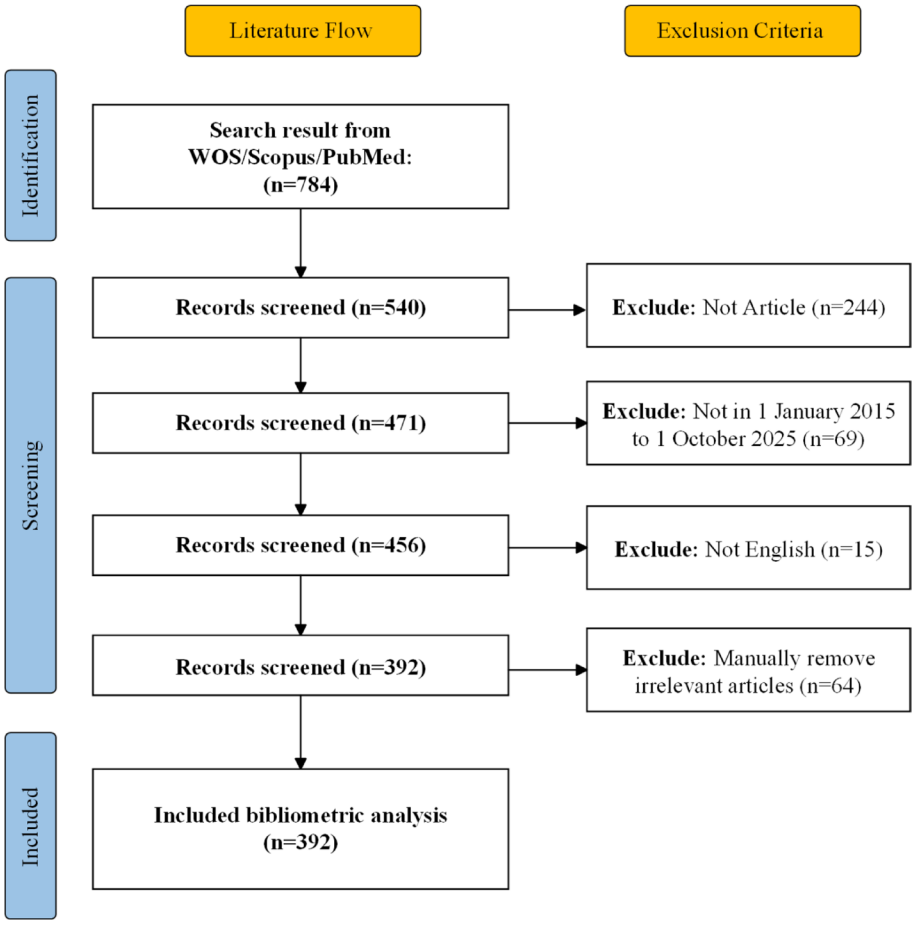


Fig. 1 PRISMA-style flow diagram of study identification, screening, and inclusion across the Web of Science Core Collection, Scopus, and PubMed

estimation" OR "volume estimation" OR "nutrient estimation" OR "energy estimation" OR "calorie estimation") AND ("artificial intelligence" OR "machine learning" OR "deep learning" OR "computer vision" OR "neural network*" OR "vision-language model*").

The search was restricted to English-language journal articles. Conference proceedings, editorials, notes, and other non-article document types were excluded at the screening stage. The complete search results from the three databases were then combined for subsequent screening and selection.

3.3 Eligibility criteria

Studies were included if they met all of the following criteria:

(1) They were published as journal articles in English;
(2) They were published between 1 January 2015 and 1 October 2025;
(3) They focused on food recognition, food classification, food detection, image-based dietary assessment, portion or volume estimation, nutrient or energy estimation, or closely related AI-supported food computing tasks;
(4) They provided sufficient methodological or empirical detail to support bibliometric analysis or narrative synthesis.

Studies were excluded if they met any of the following conditions:

(1) They were not journal articles, such as conference papers, editorials, notes, or meeting abstracts;
(2) They fell outside the predefined publication period;
(3) They were not written in English;
(4) They were clearly irrelevant to the scope of this review after title, abstract, or full-text assessment. In the narrative synthesis, priority was given to studies that reported identifiable tasks, datasets, models, and evaluation metrics. A limited number of technically informative cross-domain studies were retained only when they offered methods directly relevant to nutrient estimation, image-based food analysis, or transferable modeling strategies for dietary assessment.

3.4 Screening and study selection

The study selection process followed a PRISMA-style workflow. A total of 784 records were initially identified from the three databases. Screening was then conducted in several stages. First, non-article document types were removed, leaving 540 records. Second, studies outside the target time window were excluded, resulting in 471 records. Third, non-English records were removed, leaving 456 records. Finally, the remaining studies underwent relevance screening, during which 64 off-topic articles were excluded. The final bibliometric corpus comprised 392 articles.

To improve the reliability of the selection process, two graduate researchers independently screened the records. Titles and abstracts were reviewed first, followed by full-text checking for studies that were potentially relevant or unclear at the initial screening stage. Any disagreements were resolved through discussion, and unresolved cases were adjudicated by the corresponding author. The final set of included studies was then used for bibliometric mapping and subsequent narrative synthesis.

3.5 Data extraction and study classification

For the included studies, we extracted core bibliographic and methodological information relevant to the aims of this review, including author, publication year, study focus, task type, dataset, model or algorithm, and main evaluation metrics. For studies discussed in the narrative synthesis, we further noted information related to food recognition, portion or volume estimation, nutrient or energy estimation, and other technically relevant food-computing tasks. To support structured synthesis, the included studies were organized into several major categories, including food recognition and classification, portion or volume estimation, nutrient or energy estimation, and related image- or spectral-based food analysis. This classification helped summarize the literature from both technical and application-oriented perspectives.

3.6 Data analysis and visualization

We analyzed and visualized the corpus with the bibliometrix R package, following a standard science-mapping workflow that combines descriptive performance indicators (annual production, sources, countries, and authors) with network-based analyses. Co-occurrence matrices were normalized using the association strength measure to reduce size bias and improve cluster interpretability [11], and clusters/themes were summarized via strategic/thematic maps using Callon centrality and density [12]; longitudinal shifts were inspected with thematic-evolution views [13]. All charts and maps were generated

in bibliometrix/Biblioshiny with consistent thresholds (e.g., minimum occurrences for inclusion) and exported figures were used in the manuscript [14].

4 Result

4.1 Global trends in publications

From 2015 to 2018 the field grew slowly, fluctuating between 13 and 14 papers per year; output then more than doubled in 2019 (29) and continued a steady rise through 2020 (30), 2021 (38), 2022 (47), and 2023 (54). A pronounced surge occurred in 2024, reaching a peak of 83 articles, indicating rapid consolidation and expanding interest in this research area. The apparent decrease in 2025 (56) does not represent a downturn but reflects that the search window closed on 1 October 2025, so the count is partial-year only. Overall, the trajectory demonstrates sustained and accelerating growth since 2019, with 2024 marking the highest recorded annual output in the corpus (shown in Fig. 2).

4.2 Keyword landscape and author–topic linkages

The word-cloud highlights the core vocabulary of the corpus (Fig. 3a). The most frequent terms are mainly domain-oriented, led by dietary assessment, food, nutrition, validity/validation, health, children, and obesity. Demographic and measurement-related terms, such as adults, adolescents, energy, quality, and accuracy, are also prominent. Methodological terms, including deep learning, machine learning, artificial intelligence, image-based dietary assessment, and food recognition, appear consistently but remain secondary to nutrition and assessment terminology. Overall, the keyword profile indicates that the literature is anchored in dietary assessment and validation, while increasingly incorporating AI- and vision-based methods.

The three-fields map further shows a concentrated flow from a relatively small set of cited references to a core group of authors and then to a limited set of high-frequency topics (Fig. 3b). The main inflows come from three sources: computer-vision and

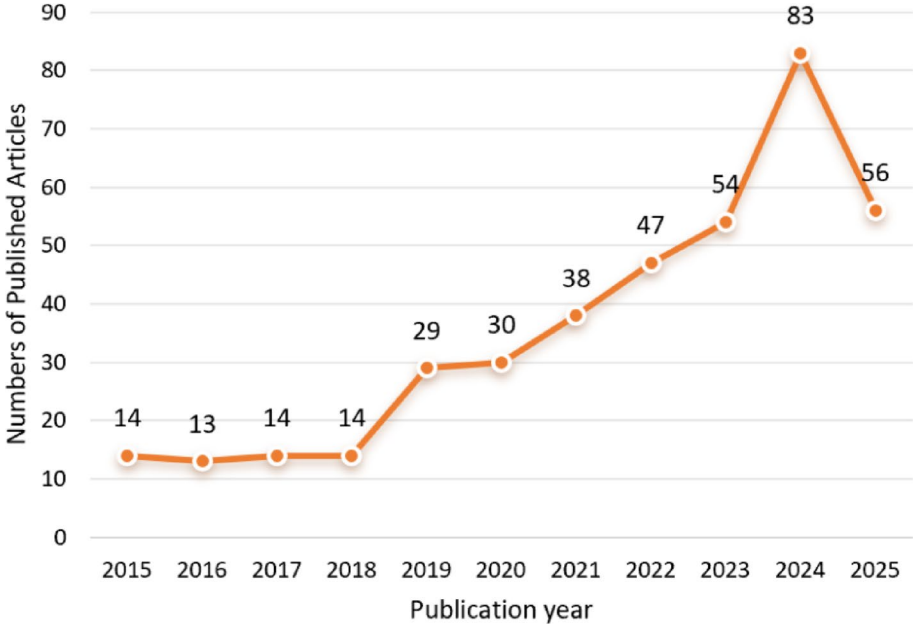


Fig. 2 Annual publications on AI-based food recognition and nutrient estimation (Web of Science Core Collection, 2015–2025; 2025 counted through October)

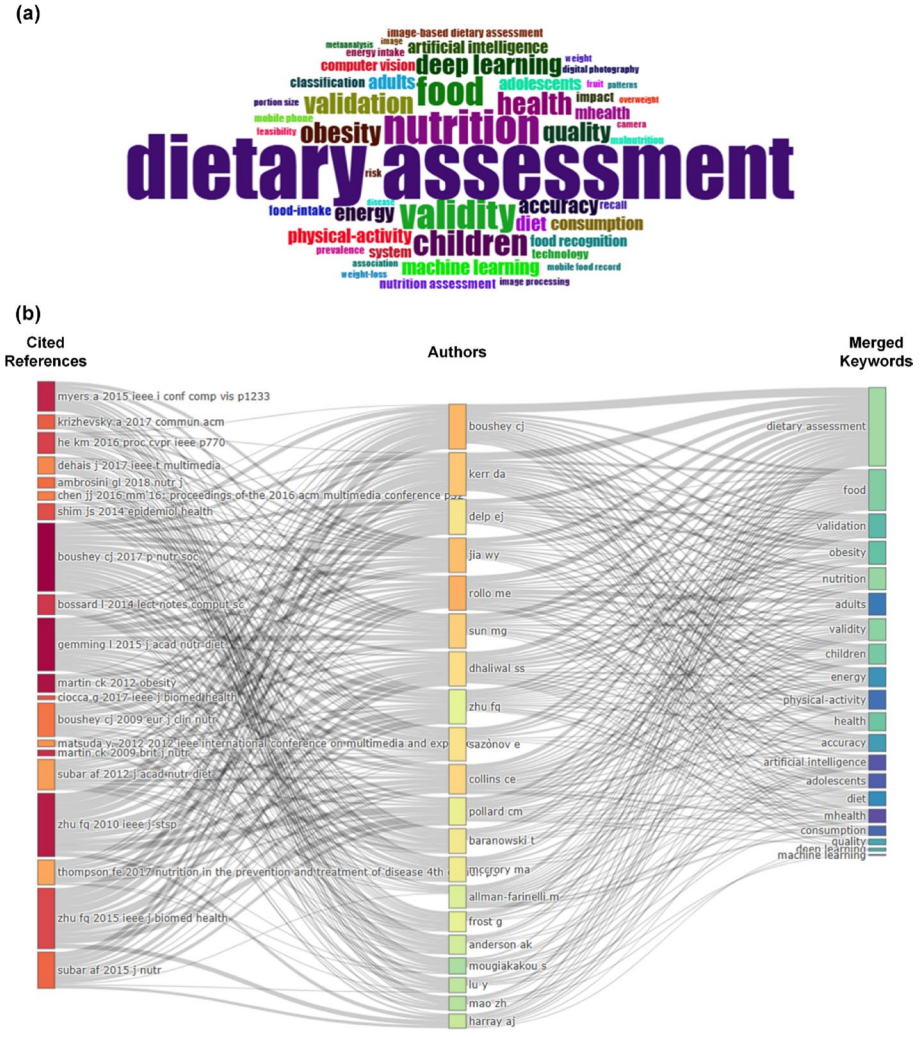


Fig. 3 Conceptual structure of the dietary-assessment literature. **a** Keyword landscape: Word-cloud of author keywords; size is proportional to term frequency across the corpus. **b** Three-fields map: Sankey-style plot linking cited references → authors → merged keywords; flow width denotes co-occurrence strength

deep-learning foundations, image-based intake and food-recognition systems, and nutrition-methodology studies on dietary assessment and validation. These streams converge on central authors such as Boushey, Kerr, Delp, Jia, Rollo, Sun, and Zhu, whose work connects both technical and nutrition-methods traditions. On the topic side, the largest outflows converge on dietary assessment, validation, food, and nutrition, with smaller but visible links to obesity, children, adults, energy, and accuracy. Overall, the plot suggests a hub-and-spoke structure in which a small core of references and authors anchors the field, while dietary assessment and validation remain the dominant thematic endpoints.

4.3 Trends and future directions in the domain

A term–time bubble timeline summarizes the temporal evolution of high-frequency keywords in the corpus (Fig. 4). Early activity is dominated by image-based dietary records and camera, reflecting an initial focus on image capture and logging. Between 2018 and 2022, the vocabulary broadens to include physical activity and obesity, indicating

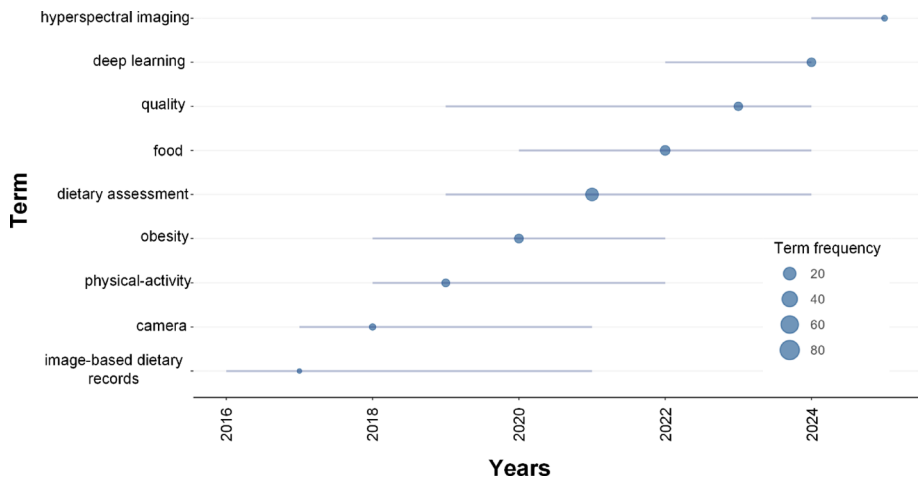


Fig. 4 Temporal evolution of high-frequency keywords in the dietary-assessment corpus (2016–2024)

increasing attention to behavioral and health-related applications. From 2020 onward, domain-centered terms such as food and dietary assessment become especially prominent, while quality spans both the mid and later periods. In the most recent interval, methodological terms become more visible, with deep learning peaking strongly in 2024 and hyperspectral imaging emerging later at lower frequency. Overall, the timeline suggests a progression from image capture and logging, to health-related application terms, and then to more explicit AI-driven methodological vocabulary.

The conceptual structure map derived from multiple correspondence analysis resolves the keyword space into two principal dimensions (Fig. 5a; Dim 1 = 38.15%, Dim 2 = 20.17%). Dimension 1 mainly separates clinical assessment language from computational language. The right side is dominated by engineering and computer-vision terms such as artificial intelligence, deep learning, classification, and image processing, whereas the left side is characterized by nutrition-methods terms such as digital photography, mobile food record, validity, and reliability. Dimension 2 further separates measurement constructs, such as nutrition assessment, accuracy, and portion size, from population and implementation terms, including adolescents, obesity, risk, patterns, technology, and mobile food record.

The thematic map supports this interpretation by positioning clusters according to relevance and internal development (Fig. 5b). Two motor themes occupy the upper-right quadrant: dietary assessment–nutrition–validity and obesity–physical activity–adolescents. The lower-right quadrant contains basic but less internally developed themes, including food–quality–machine learning and classification–image processing–hyperspectral imaging. Spectroscopy-related terms appear as niche themes in the upper-left quadrant, while diagnosis and adult patients fall into the emerging or declining quadrant. Overall, the map shows that the field is centered on dietary assessment and validation, while AI-, vision-, and processing-related themes are increasingly integrated into this core structure.

Overall, the map shows that the field is centered on dietary assessment and validation, while AI-, vision-, and processing-related themes are increasingly integrated into this core structure. Taken together, these bibliometric patterns show that the field is anchored in dietary assessment, nutrition, and validity, while AI-driven food analysis, obesity-related applications, and child-relevant themes form increasingly visible

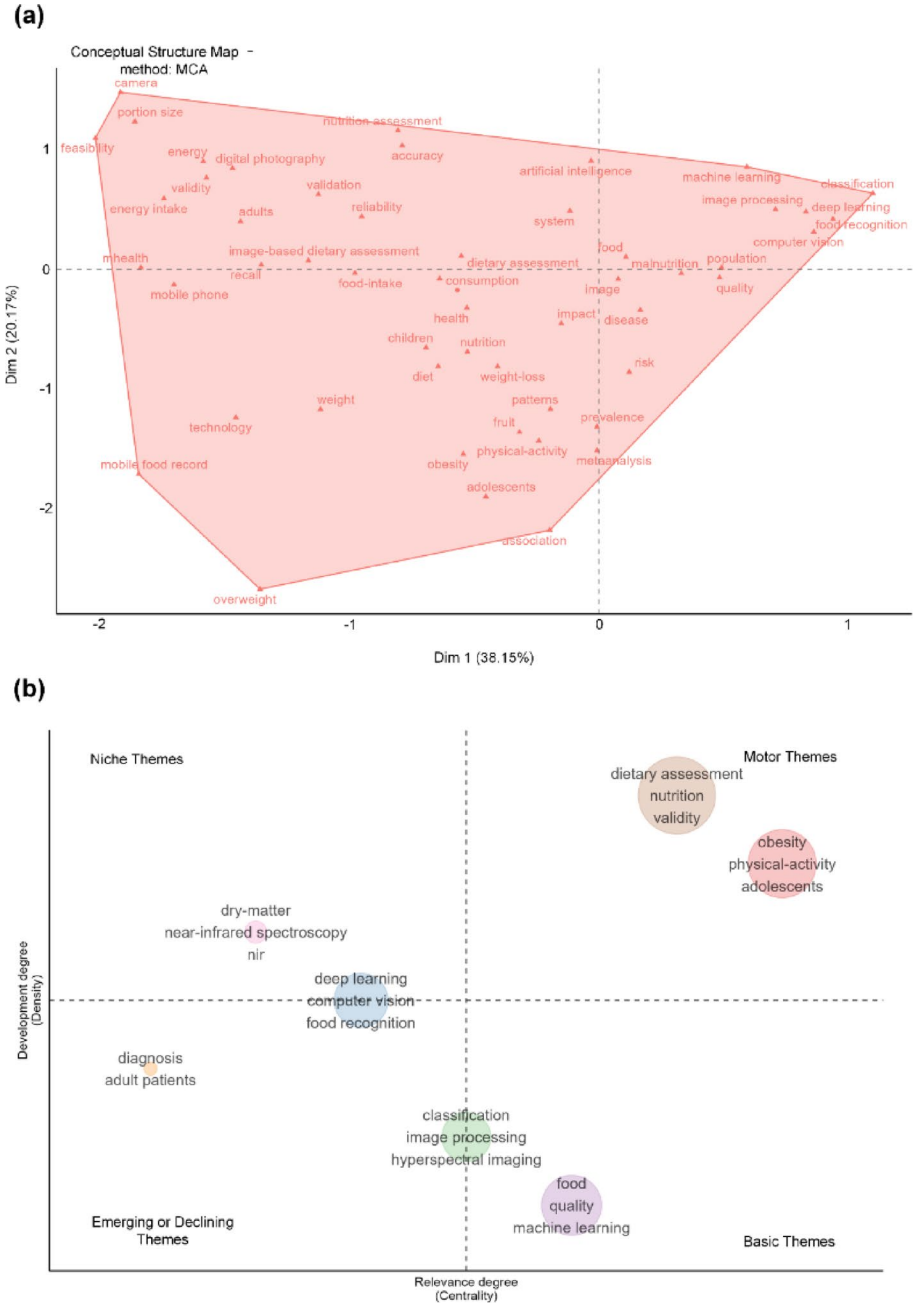


Fig. 5 Conceptual and strategic mapping of research themes. **a** Conceptual structure map (MCA): Two-dimensional projection of author keywords obtained by multiple correspondence analysis (Dim 1 = 38.15%; Dim 2 = 20.17%). **b** Thematic map (centrality–density): Strategic diagram plotting relevance (centrality) versus development (density), partitioning topics into Motor, Basic, Niche, and Emerging/Declining themes; bubble size reflects term frequency

connected strands. Guided by this structure, the narrative synthesis that follows focuses on four task domains with the clearest translational relevance: food recognition and segmentation, portion and volume estimation, nutrient estimation, and pediatric/free-living validation.

4.4 Empirical evidence across core IBDA tasks

To present the collected studies more clearly and to strengthen the connection between the bibliometric mapping and the narrative review, the included literature is organized here by task rather than listed in a flat sequence. These task-oriented sections were derived from the dominant themes and translational fault lines identified in the science-mapping results. In particular, the preceding analyses showed a field centered on dietary assessment, nutrition, and validity, while repeatedly linking these themes to food recognition, AI-driven analysis, obesity, and child-relevant application contexts. On this basis, the following synthesis focuses on four domains that are both technically central and practically relevant: food recognition and segmentation, portion and volume estimation, nutrient estimation, and pediatric/free-living validation. Table 1 summarizes the main characteristics of representative core studies, Table 2 synthesizes the major AI techniques and their representative applications, and Table 3 contrasts benchmark performance with evidence from more naturalistic or real-world settings. This structure makes it easier to distinguish methodological maturity from translational readiness.

4.4.1 Food recognition and segmentation studies

Food recognition is the most mature component of image-based dietary assessment. The collected studies cover datasets that range from relatively small targeted collections to large benchmark corpora. Representative examples include a Chinese food dataset with 8,734 images across 25 categories [14], the Central Asian Food Dataset with more than 16,000 images across 42 classes [32], a Food-101–derived corpus with approximately 100,000 images across 100 classes [12], and the NutriNet recognition and detection corpora with 225,953 and 130,517 images, respectively [27]. These studies show that the field already has a substantial dataset base for benchmarking recognition models.

Recognition performance is generally strong under curated conditions. A compact five-layer CNN achieved Top-1 accuracy of 97.12% and Top-5 accuracy of 99.86% on the Chinese food dataset [14]. ResNet-152 reached Top-1 accuracy of 88.70% on CAFD [32], while NutriNet reported Top-1 accuracy of 86.72% for recognition and 94.47% detection accuracy on a dedicated detection set [27]. On a Mediterranean-food dataset, EfficientNet-B2 achieved Top-1 accuracy of 83.8% and Top-5 accuracy of 97.6%, and a CLIP-based zero-shot recognizer reached Top-1 accuracy of 91% on CNFOOD-241 [23]. These results indicate that CNN, transfer-learning, and more recent vision-language approaches can all provide strong recognition performance when images are relatively standardized and label definitions are clear.

Segmentation and detection are also important because they support downstream portion estimation and nutrient inference. Multiple-hypothesis image segmentation combined with a feed-forward neural classifier reported 96% classification accuracy before calorie lookup [13]. In multi-dish settings, EfficientDet achieved mAP = 0.92 across 87 dish types [26]. NutriNet also combined recognition with FCN- and ResNet-based segmentation, showing that food recognition is increasingly being embedded within multi-stage dietary-assessment pipelines rather than treated as a stand-alone classification problem [27].

Table 1 Characteristics of included core studies

References	Setting/population	Modality	Task	Dataset/size	Model	Metric	Pediatric relevance
[15]	Benchmark; 20 patients/60 dishes	RGB-D	BU/volume proxy	Diabetes60; ~ 9k images	FCN + BU regression	RMSE 1.53 BUs	Indirect
[16]	Controlled images; general foods	RGB	Segmentation + calorie	Segmented food images	Multiple-hypothesis segmentation + FFNN	Cls acc 96%	Indirect
[17]	Benchmark; general foods	RGB	Recognition + nutrient cues	Food-101 ext.; ~ 100k/100 classes	Deep CNN pipeline	Top-1 up to 85%	Indirect
[18]	Benchmark; Chinese foods	RGB	Recognition	8,734 images/25 categories	5-layer CNN	Top-1 97.12%; Top-5 99.86%	Indirect
[19]	Controlled experiment; multiple foods	Depth/video	Portion estimation	Single-depth or video samples	Point completion + view synthesis	Improved over prior baselines	Indirect
[20]	Crowdsourced restaurant meals	RGB	Nutrient estimation	15,908 images/470 restaurants	Commercial DL model	Acc 75.1% vs manual coding	Indirect
[21]	Controlled feasibility study; 101 foods	RGB + pred. UV/NIR	Recognition + calorie	101-food dataset	RGB-to-multispectral DNN	Calorie MAPE 12.13%; recognition 98.24%	Indirect
[22]	Benchmark; long-tail foods	RGB	Recognition	Food101-LT; VFN-LT	Single-stage heavy-tailed framework	Top-1 improvement > 5%	Indirect
[23]	Benchmark; general meals	RGB	Energy via 3D reconstruction	Nutrition5k	Monocular voxel reconstruction	MAE 40.05 kcal; MAPE 11.47%	Indirect
[24]	Benchmark; Central Asian foods	RGB	Recognition	CAFD; > 16k images/42 classes	ResNet-152	Top-1 88.70%	Indirect
[25]	Benchmark; general foods	RGB	Detection + portion	FPB; 14,083 images/138 classes	YOLOv12 multitask model	mAP50 0.978; MAE 90.95 g	Indirect
[26]	Benchmark; Chinese dishes	RGB	Recognition + nutrient	CNFOOD-241	RegNetY ensemble	Top-1 86.28%; mean R2-top5 0.8636	Indirect
[27]	Benchmark + advisory system	RGB/multi-view	Recognition + volume	CNFOOD-241; Food-101	CLIP + MVS + LLM	Top-1 91%; volume MAPE 23.5%	Indirect
[28]	Controlled multi-dish setting	RGB	Detection/recognition	87 dish types	EfficientDet	mAP 0.92	Indirect
[29]	Benchmark + smartphone test	RGB	Recognition + segmentation	225,953 recog.; 130,517 detect	NutriNet + FCN/ResNet	Top-1 86.72%; real-world Top-5 55%	Indirect

Pediatric relevance was coded conservatively. Direct pediatric evidence was defined as validation involving children or adolescents in child-centered deployment settings, such as school cafeterias, household or family-based intake monitoring, or pediatric obesity-management workflows. Studies were coded as indirect when they demonstrated technical capability for recognition, portion estimation, or nutrient estimation but did not evaluate performance, usability, or validity in pediatric settings. Therefore, "Indirect" indicates potential relevance to pediatric dietary assessment rather than direct pediatric deployment evidence

Table 2 Main AI techniques and representative applications

Technique	Typical models	Input modality	Output	Strengths	Limitations	Representative studies
Food recognition	CNN, ResNet, RegNetY	RGB	Food class labels	Strong benchmark accuracy; mature task	Sensitive to mixed dishes and long-tail classes	[17, 18, 22, 24, 26, 29]
Segmentation/detection	FCN, ResNet seg., EfficientDet	RGB	Food regions/item localization	Supports downstream quantity and nutrient estimation	More difficult in cluttered real-world images	[16, 28, 29]
Portion/volume estimation	View synthesis, voxel reconstruction, YOLOv12 multitask	RGB, depth, multi-view	Portion size/volume/BU proxy	Moves toward quantitative intake estimation	More sensitive to angle, plating style, and image quality	[15, 19, 23, 25, 27]
Nutrient/energy estimation	Database linkage, ensembles, direct regression	RGB	Calories/nutrients/energy	Most clinically meaningful output	Depends on recognition and portion accuracy	[20, 21, 26, 27]
Vision-language integration	CLIP-based multimodal systems	RGB/multi-view	Recognition + advisory outputs	Better semantic flexibility; zero-shot potential	Limited real-world validation	[27]
Spectral enhancement/proxies	RGB-to-UV/NIR, MSI, transmittance imaging	Spectral or RGB-derived spectral features	Proxy nutrient traits/calorie-related cues	Captures signals not easily visible in RGB	Mostly tested in controlled settings	[21, 30, 31]

Spectral studies are included here as technically informative methods that may support dietary assessment, rather than as direct evidence of mature clinical or pediatric IBDA deployment

Table 3 Comparison between benchmark and real-world studies

Study type	Representative studies	Dataset condition	Performance	Main limitation	Clinical/pediatric implication
Curated recognition benchmark	[17, 18, 24, 26, 29]	Standardized food images with defined classes	Top-1 commonly about 85% to 97%	May overestimate practical performance	Useful for method development, but not sufficient for deployment claims
Multi-dish detection benchmark	[25, 28]	Controlled dish images with multiple items	Detection mAP about 0.92 to 0.978	Still less complex than free-living meal capture	Valuable for building pipeline front ends
3D/portion benchmark	[15, 19, 23, 25, 27]	Lab-measured weights or geometric targets	MAPE about 11% to 12%; portion MAE about 90.95 g	Sensitive to image quality and dish geometry	Important for intake estimation, but not yet robust enough for routine real-world use
Crowdsourced restaurant images	[20]	Naturalistic but still relatively structured user images	Accuracy 75.1% vs manual coding	Food presentation and coding variability remain high	Suggests practical promise, but still limited ecological breadth
Smartphone free-living validation	[29]	Unconstrained real-world capture	NutriNet Top-5 55%	Lighting, occlusion, mixed dishes, and user capture behavior reduce performance	Shows that benchmark success does not directly translate to routine dietary monitoring
Pediatric/child-centered application	This review synthesis	Very limited direct evidence	Mostly indirect relevance only	Few studies conducted in school, family, or pediatric obesity settings	Pediatric deployment remains an evidence gap rather than an established use case

4.4.2 Portion and volume estimation studies

Portion and volume estimation remain more difficult than recognition, but the reviewed studies show clear technical progress. Current approaches include single-image 3D reconstruction, stereo vision, depth-assisted estimation, and multitask detection-portion models. The Nutrition5k study used monocular voxel reconstruction and reported MAE = 40.05 kcal and MAPE = 11.47% for energy estimation from a single RGB image [20]. The Food Portion Benchmark combined detection and portion estimation in a single YOLOv12-based model and achieved mAP50 = 0.978 for detection and MAE = 90.95 g for portion estimation [21].

Other studies show that geometric information remains important when estimating intake quantity. A stereo-image pipeline for Mediterranean foods reported a volume MAPE of 10.5%. A view-synthesis method based on single depth image or video improved portion estimates relative to earlier baselines [16]. In Diabetes60, Bread Units were estimated by predicting depth from RGB and then regressing BUs from the inferred depth, reaching RMSE = 1.53 BUs, compared with human rater variability of RMSE = 0.89 [33]. These results suggest that portion estimation has moved beyond simple heuristic assumptions, but it is still more sensitive to imaging conditions and meal presentation than recognition.

4.4.3 Nutrient estimation studies

Nutrient and energy estimation are the most application-relevant outputs in the IBDA pipeline, but they also require stronger integration across recognition, quantity estimation, and food-composition mapping. The reviewed studies show several technical routes, including direct prediction from images, recognition followed by database lookup, RGB-to-multispectral estimation, and multimodal systems that combine recognition with geometric reconstruction.

Results are promising but still variable across settings. A commercial deep-learning system applied to 15,908 crowdsourced restaurant-food images from 470 restaurants achieved 75.1% accuracy when compared with manual coding [34]. On CNFOOD-241, a RegNetY-based ensemble reached Top-1 accuracy of 86.28%, Top-5 accuracy of 98.49%, and mean R^2-top5 of 0.8636 for nutrient estimation [22]. A multimodal advisory system that combined CLIP recognition with learning-based multi-view stereo reported Top-1 accuracy of 91% and volume MAPE of 23.5% [23]. In another study, RGB images were first converted into predicted UV/NIR representations, resulting in recognition accuracy of 98.24% with predicted multispectral images and 99.45% with actual multispectral images, while calorie MAPE was 12.13% across 101 foods [35].

Some technically related studies also suggest that spectral or proxy-based image signals may help capture nutritional characteristics that are difficult to infer from RGB appearance alone. Oil detection on salad leaves using 10-band multispectral imaging achieved 84.2% accuracy [24], and a transmittance imaging approach with a deep autoencoder classified vitamin-A–related puree stages at 80% accuracy [25]. These studies are not full end-to-end dietary-assessment systems, but they support the idea that spectral information can complement conventional food-image analysis.

4.4.4 Pediatric and free-living validation studies

A central translational issue is whether strong benchmark performance can be maintained in free-living settings. The reviewed literature suggests that this remains a major challenge. NutriNet achieved strong results on curated datasets, but in a real-world smartphone test its Top-5 accuracy dropped to 55% [27]. This gap illustrates the performance loss that can occur when models are exposed to unconstrained lighting, occlusion, mixed dishes, packaging, and variable shooting angles. To make this contrast clearer, Table 3 now links each benchmark or real-world evidence category to representative studies and summarizes the corresponding performance range. These comparisons show that high benchmark performance does not necessarily translate into equivalent accuracy under unconstrained food-capture conditions.

Several studies provide partial evidence for practical use, although explicitly pediatric validation is still limited. The crowdsourced restaurant-food study [34] reflects a more naturalistic image source than laboratory datasets, while CNFOOD-241 and region-specific food datasets are closer to real meal settings than highly standardized benchmarks [22, 23, 32]. Psychophysical findings further show that humans can rapidly infer energy density and macronutrient cues from food images, with significant correlations between estimated and true values even at short exposure times [17]. In the 15 core studies summarized in Table 1, no study directly validated an IBDA system in a child-centered deployment setting. All 15 studies were therefore coded as having indirect pediatric relevance, with evidence derived mainly from curated benchmarks, controlled or semi-controlled imaging studies, crowdsourced restaurant images, and one smartphone-based free-living test.

Overall, the evidence across these task groups shows a consistent pattern. Food recognition is the most mature area, portion and volume estimation are improving but remain more sensitive to imaging conditions, nutrient estimation is increasingly feasible through integrated pipelines, and real-world or pediatric validation is still the least developed part of the field.

4.5 Main AI techniques and representative applications

Image-based dietary assessment is supported by several recurring technical routes. The first is RGB-based food recognition, which is dominated by CNNs, transfer-learning models, and, more recently, vision-language models [12, 14, 22, 23, 27, 32]. These methods perform strongly on curated datasets and are now widely used as the front end of dietary-assessment systems. The second is segmentation and detection, which localize food items before downstream estimation and are especially important for mixed dishes and multi-item meal images [13, 21, 26, 27]. The third is portion and volume estimation, where current solutions rely on stereo geometry, inferred depth, 3D reconstruction, or multitask learning to convert image appearance into quantitative intake measures [16, 20, 21, 23, 33]. The fourth is nutrient inference, which either maps recognized foods to composition databases or predicts nutrient-related properties more directly from image-derived features [22, 23, 34, 35].

From a translational perspective, these technical routes differ not only in maturity but also in practical value. Recognition and detection are already relatively mature under benchmark conditions and can provide the basic front end for dietary-monitoring workflows. Portion and volume estimation move the field closer to actual intake

measurement, but performance remains more sensitive to image quality, dish geometry, and user behavior. Nutrient inference is the most relevant output for public-health and obesity-related applications because it connects food images to calories, nutrients, and dietary patterns. However, it is also the most dependent on the quality of upstream recognition and quantity estimation. For this reason, the technical performance of an end-to-end system should not be judged solely by classification accuracy.

In addition to these core IBDA techniques, a smaller group of image- and spectral-based studies from adjacent food-analysis domains was reviewed because they illustrate transferable technical ideas rather than serving as direct dietary-assessment evidence. These studies include multispectral and hyperspectral approaches for oil detection, moisture prediction, pigment mapping, fraud detection, freshness inspection, and composition-related proxy estimation [24, 25, 28–31, 36–43]. Their main value in the present review is methodological: they show that spectral modality, band selection, and spatial-spectral fusion can improve prediction when RGB information alone is insufficient. For this reason, these studies should be interpreted as supporting evidence for technical transfer, not as core validation studies of image-based dietary assessment in clinical or pediatric settings.

Taken together, the reviewed literature shows that the technical center of the field lies in recognition, segmentation, portion estimation, and nutrient mapping, while spectral methods currently play a complementary role by extending what can be inferred from visual signals under controlled conditions. For a public-health audience, the key issue is not whether increasingly complex models can achieve higher benchmark scores, but whether these methods can support reliable, low-burden, and interpretable dietary assessment in real-world settings, especially for populations relevant to obesity prevention and care (Table 4).

5 Discussion

5.1 From benchmark performance to real-world readiness

This review shows that AI-enabled image-based dietary assessment has made clear technical progress, but the distance between benchmark success and real-world readiness remains substantial. Across the reviewed studies, food recognition is the most mature stage of the pipeline, and recent models often achieve strong performance on curated datasets. Portion and volume estimation have also improved, and nutrient inference is becoming increasingly feasible through integrated recognition-and-estimation pipelines. These developments indicate that the field has moved beyond early proof-of-concept work and is now capable of supporting more realistic forms of dietary measurement.

At the same time, the strongest published results are still concentrated in relatively controlled settings. Benchmark datasets usually provide cleaner images, narrower label spaces, and more standardized viewing conditions than those seen in everyday food logging. Once systems are exposed to free-living environments, performance becomes less stable because errors accumulate across recognition, segmentation, portion estimation, and nutrient mapping. Lighting variation, mixed dishes, occlusion, packaging, region-specific foods, and user capture behavior can all reduce reliability. For this reason, benchmark performance should be interpreted as evidence of technical potential rather than direct proof of clinical or public-health readiness.

Table 4 Adjacent image- and spectral-based studies included as method-transfer evidence for food analysis and dietary-assessment modeling

References	Task	Data (Dataset, Size)	Approach/Model	Metric(s)
[36]	Nutrient	Macadamia leaves; HSI 400–1000 nm; 30 trees; multiple timepoints	PLSR; adaxial vs. abaxial surface comparison; RPD evaluation	N: R^2 0.55, RPD 1.52; P: R^2 0.77, RPD 2.11; K: R^2 0.77, RPD 2.12; Ca: R^2 0.75, RPD 2.04
[30]	Recognition (oil detection in salads)	Custom multispectral salad-leaf dataset; 10 bands (380–980 nm)	Pseudo-reflectance + ANOVA/Tukey + SFS wavelength selection	Accuracy 84.20%
[37]	Nutrient (moisture)	Macadamia nuts (in-shell & kernels); selected HSI bands 760–967 nm	PLSR, ANN, SVM, GPR; auto/manual spectra extraction; orientation study	In-shell: R^2T 0.96, RMSET 1.20%, RPD 5.15; Kernels: R^2T 0.99, RMSET 0.308%, RPD 11.05
[38]	Nutrient	17 edible algae species	SWIR HSI + ANN multi-species models	R^2: proteins 0.9952; lipids 0.9767; fibers 0.9828
[39]	Recognition (food fraud)	Turmeric powder images: 6,240; 13 categories	Improved CNN with gated pooling (avg + max); preprocessing + augmentation	Accuracy 99.36%
[40]	Nutrient (moisture)	Edamame; HSI reflectance + transmittance	Fusion spectra + chemometrics (PLSR/LSSVR with STA/SNV)	Best (intermediate fusion, SNV + LSSVR): RP 0.9149, RPD 2.4
[41]	Nutrient (SSC)	Dried Hami jujube; SWIR HSI 1000–2500 nm; positions compared	Custom CNN vs. PLSR vs. SVR; CARS/GA/IRIV variable selection	CNN (full spectra): R^2p 0.857, RMSEP 0.563, RPD 2.648
[44]	Recognition (egg freshness/defects)	Eggs: 1,000 fresh + 800 defective; HSI + band selection	DNN/1D-CNN/2D-CNN/3D-CNN/MobileNet; real-time system	Freshness acc 99%; defect acc 100%; real-time defect 95% @ 31 ms
[42]	Recognition (egg quality via shape)	Egg images: 6,378	Image analysis features; multiple ML models; Random Forest best	Accuracy 97.9%
[43]	Recognition (geographical origin)	Chrysanthemum HSI; size n/r	CIMPSO wavelength selection + spatial texture/color; RF & others	CIMPSO: Acc 94.16–96.02%; fused spatial-spectral RF: Acc 98.86%, F1 98.84%
[45]	Nutrient (pigments/phenolics)	Lettuce (11 cultivars); Headwall HSI + MicaSense multispectral	PLSR & Random Forest; spatial mapping	RF: R^2 0.91 (anthocyanins), 0.76 (flavonoids); phenolics R^2 0.79
[46]	Recognition (species authentication)	Porcini mushrooms; FT-IR + GC–MS; 773 metabolites	2DCOS image recognition; network pharmacology & docking	-
[47]	Recognition (beef quality)	Yunling snowflake beef: 250 samples; HSI 900–2500 nm	Transformer-based autoencoder (3 layers) + CatBoost classifier	Average accuracy 82.42%
[48]	Volume + Nutrient (composition proxy)	Pork neck muscles from 20 carcasses; 3-D scans (point clouds)	3-D volume → density; correlations with lab composition	r: water 0.5213; protein 0.5887; fat − 0.6601 ($p < 0.05$)
[49]	Recognition (UAV land-cover)	UAV multispectral (NIR,R,G,B); GSD 0.0125 m	OBIA (KNN) vs PBIA (ML); pixel vs object-based comparison	OBIA OA 94.9%; Kappa 93.4% (OBIA > PBIA by 21% OA)

This distinction is especially important for obesity-related applications. In practice, a usable system must do more than classify food images accurately. It must also operate under ordinary conditions, tolerate imperfect images, and provide outputs that remain meaningful when uncertainty is unavoidable. From a public-health perspective, the central question is therefore not whether a model can achieve another incremental gain on a curated benchmark, but whether it can support dependable, low-burden dietary assessment in real settings.

5.2 Comparison with previous review papers

Compared with earlier review papers in this field, the present review offers a broader translational perspective. Previous reviews, such as Tahir and Loo's survey of image-based food recognition and volume estimation methods [4], Dalakleidi et al.'s systematic review of image-based food-recognition systems for dietary assessment [5], and Lo et al.'s review of image-based food classification and volume estimation for dietary assessment [34], mainly emphasized technical approaches, task definitions, and model development. These works made an important contribution by clarifying the methodological foundations of the field and summarizing the main recognition and estimation strategies.

The present review extends that foundation in several ways. First, it combines bibliometric mapping with narrative synthesis, which allows the field to be examined at both the macro level and the study level. Second, it updates the evidence base through 2025 and incorporates more recent developments in multimodal learning, spectral augmentation, and end-to-end dietary-assessment pipelines. Third, it places stronger emphasis on the contrast between benchmark performance and real-world applicability. Most importantly, this review uses pediatric obesity prevention and care as an interpretive lens, rather than treating childhood relevance as a brief downstream application. This framing helps shift the discussion from technical possibility alone to translational suitability, which is a key issue for public-health deployment.

5.3 Public-health relevance for pediatric obesity prevention and care

The pediatric obesity focus is important because children's dietary assessment poses challenges that are not fully captured in adult-centered or laboratory-based workflows. Eating occasions in childhood are often irregular, partially supervised, and frequently reported by caregivers rather than by the child directly. School meals, snacks, beverages, shared foods, and packaged items all complicate image capture and portion estimation. These features make pediatric dietary assessment both more difficult and more relevant to public-health intervention.

From this perspective, the value of IBDA lies not only in reducing reporting burden but also in making dietary monitoring more continuous, timely, and behaviorally meaningful. For pediatric obesity prevention and care, the most useful outputs may not always be exact calorie values for every meal. In many cases, greater value may come from stable meal-pattern indicators, such as eating frequency, timing regularity, beverage consumption, snack exposure, and repeated appearance of high-risk food categories. These types of signals are often more actionable for clinicians, families, and schools than highly precise but less robust nutrient estimates.

The current literature suggests that the technical base for such applications is emerging, but direct pediatric validation is still limited. Many reviewed studies are relevant to

pediatric use only indirectly, because they demonstrate recognition, portion, or nutrient-estimation capability without testing systems in child-centered settings. As a result, the pediatric obesity perspective should not be treated as a simple extension of adult dietary assessment. It requires its own validation logic, its own deployment settings, and its own standards for practical usefulness.

5.4 Ethical considerations and pediatric-specific validation needs

Ethical considerations become more important when dietary assessment systems are designed for children. Food images collected in homes, schools, or clinical programs may contain not only meal information but also contextual details about daily routines, family habits, and socioeconomic conditions. When minors are involved, privacy protection should therefore be treated as a central design requirement rather than a secondary technical feature. Data minimization, secure storage, restricted sharing, and privacy-preserving processing strategies are especially important in pediatric use cases.

Consent and participation issues also differ from those in adult deployment. In pediatric settings, consent may involve parents or guardians, while actual image capture may depend on children, caregivers, or school staff. This creates practical questions about assent, proxy reporting, responsibility for image quality, and the interpretation of uncertain results. A system that is technically accurate but difficult for families to use consistently may have limited real-world value. For this reason, pediatric workflows should be designed to fit naturally into family routines and school environments rather than assuming ideal user compliance. This may include caregiver-friendly feedback, such as photographic portion exemplars, and school-compatible workflows linked to cafeteria or menu information.

Another major issue is validation. Pediatric-specific validation should go beyond general benchmark testing and should examine whether systems remain reliable across age groups, meal contexts, and social environments. In practical terms, a pediatric-specific benchmark dataset should include child and adolescent meal images collected from child-relevant settings, such as homes, school cafeterias, and family-based obesity-management contexts. It should represent pediatric food ecologies, including school meals, snacks, beverages, packaged foods, mixed dishes, shared foods, and leftovers, and should be linked to appropriate reference standards such as weighed meal records, controlled meal protocols, school menu data, or dietitian annotations. Such a dataset should also include information on portion size, meal timing, missed eating occasions, image quality, capture context, and basic subgroup variables so that future systems can be evaluated for recognition accuracy, portion and nutrient estimation error, usability, uncertainty, and fairness in pediatric settings.

Future studies should include validation in school cafeterias, household meal settings, and family-based obesity-management programs. They should also consider subgroup analyses by age, sex, cultural food environment, and socioeconomic context. Where possible, performance should be anchored against appropriate gold standards, such as weighed meal records or controlled meal protocols at the meal level, and broader study-level validation should be aligned with established nutritional assessment standards. Uncertainty reporting is also essential. In pediatric care, output should help guide better decisions, not create a false sense of precision.

5.5 Methodological implications and practical directions

Several methodological priorities follow from the present synthesis. First, robustness to long-tail foods and mixed dishes remains essential. In real dietary assessment, especially in diverse child populations, food categories are unevenly distributed and many relevant items appear infrequently. Models that rely too heavily on dominant benchmark classes may therefore perform well in publications but less well in practice. Approaches that incorporate ingredient-level reasoning, culturally specific food ontologies, or adaptive updating strategies are likely to be more useful in heterogeneous environments.

Second, portion estimation remains a key bottleneck. The literature shows that geometry-aware approaches, such as inferred depth, stereo cues, multi-view reconstruction, or joint detection-portion models, can improve quantitative estimation. However, these methods must be balanced against user burden and workflow constraints. In public-health settings, the goal is not to maximize technical complexity, but to identify solutions that produce sufficiently reliable quantity information with minimal friction.

Third, spectral and multimodal methods should be interpreted carefully. The reviewed cross-domain studies suggest that spectral information can improve prediction for certain nutrition-related proxies, especially when RGB appearance alone is insufficient. However, such methods should currently be viewed as complementary rather than universal solutions. Their main contribution at present is to show possible directions for method transfer, not to provide direct evidence of mature dietary-assessment deployment.

Overall, the field appears to be moving in the right direction, but practical impact will depend on whether future work is judged by ecological validity, usability, equity, and clinical relevance, rather than by benchmark performance alone.

6 Conclusion

This review contributes to the image-based dietary assessment literature by moving beyond a method-centered summary of food-recognition and nutrient-estimation techniques. By combining bibliometric mapping with task-oriented narrative synthesis, it identifies the field's major knowledge structure, compares benchmark performance with free-living applicability, and evaluates IBDA technologies through a pediatric obesity prevention and care lens. Compared with earlier reviews that primarily emphasized technical methods, this review highlights translational readiness, pediatric deployment gaps, and the pediatric-specific evaluation criteria needed before AI-enabled dietary assessment can be used meaningfully in child-centered public-health and clinical settings.

The evidence suggests that food recognition is the most mature component under curated conditions, whereas portion estimation and nutrient inference remain more vulnerable to image quality, camera angle, mixed dishes, recipe variability, and error propagation across the IBDA pipeline. Detection and segmentation can support downstream estimation but are still challenged by occlusion, shared plates, and cluttered meal scenes. Multimodal and spectral approaches are promising, but their practical value remains limited by insufficient validation in free-living and pediatric settings.

For practice, the most immediately useful pediatric applications may not be exact per-item calorie estimation, but more robust meal-pattern and high-leverage dietary indicators, such as eating frequency, meal timing, snack exposure, beverage consumption or

sugar-sweetened beverage exposure, and repeated high-risk food categories. These outputs are likely to be more interpretable and actionable for clinicians, dietitians, families, and schools.

Future pediatric IBDA research should therefore use dedicated evaluation criteria, including child-centered deployment settings, caregiver involvement, image-capture feasibility, missed eating occasions, pediatric food ecology, appropriate reference standards, uncertainty reporting, privacy protection, and subgroup performance. Such pediatric-first validation is necessary before AI-enabled dietary assessment can become dependable support for childhood obesity prevention and care.

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Author contributions
Tianyu Gao conceived and designed the study and supervised the research project. Ying Xiao performed the data analysis, generated the figures and tables, and contributed to interpretation of results. Tianyu Gao and Ying Xiao drafted the manuscript. All authors critically reviewed and revised the manuscript for important intellectual content and approved the final version.

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Declarations

Ethics approval and consent to participate
Not applicable. This study did not involve human participants or animals. This study did not involve human participants, and informed consent was therefore not required.

Consent for publication
Not applicable. This study does not contain any individual person's data in any form, including individual details, images, or videos.

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